Amazon Alexa Customer Review Sentiment Analysis Using Machine Learning

1. Introduction

Sentiment analysis is a subfield of Natural Language Processing (NLP) that focuses on identifying and classifying opinions expressed in text into predefined categories such as positive, negative, or neutral. In this project, sentiment analysis is applied to customer reviews of Amazon Alexa products to determine whether a review reflects a positive or negative sentiment.

The dataset used in this study contains approximately 3000 customer reviews of Amazon Alexa devices, including variations such as Echo Dot and Echo Show. Each record includes information such as product variation, rating, review text, and a sentiment label. The primary objective is to develop a machine learning model capable of accurately classifying customer feedback based on textual content.

This type of analysis is valuable for businesses as it helps understand customer satisfaction, identify product strengths and weaknesses, and support data-driven decision-making.

2. Dataset Description

The dataset used for this project is sourced from Kaggle and consists of customer reviews for Amazon Alexa products. It includes the following features:

- **rating**: Numerical rating provided by the customer on a scale of 1 to 5
- **date**: Date when the review was posted
- **variation**: Type or model of the Alexa product
- **verified_reviews**: Raw textual review written by the customer
- **feedback**: Target variable indicating sentiment (1 = positive, 0 = negative)

For this study, the feedback column is used as the dependent variable, while the remaining features are used for prediction after preprocessing.

3. Data Preprocessing

Data preprocessing is a critical step in preparing raw textual data for machine learning models. The following preprocessing steps were applied:

1. **Removal of Irrelevant Features**
   The date and rating columns were removed as they do not directly contribute to sentiment classification.
2. **Handling Categorical Variables**
   The variation column was transformed using **one-hot encoding**, converting categorical values into binary numeric features.
3. **Text Cleaning and Transformation**
   The verified_reviews column contains unstructured text data. This was processed using the

CountVectorizer method, which converts text into a matrix of token counts while removing common stopwords.

4. **Feature Consolidation**
   The encoded categorical features and text-based numerical features were combined into a single dataset for model training.

These steps ensured that the dataset was fully numeric and suitable for machine learning algorithms.

4. Feature Engineering

Feature engineering involved transforming raw text into a structured format that machine learning models can interpret.

The **Bag of Words (BoW)** approach was used via CountVectorizer. In this method, each review is represented as a vector based on word frequency. Common words such as “the”, “is”, and “and” are removed, while meaningful words contribute to the feature space.

This approach allows the model to learn patterns in word usage associated with positive or negative sentiment. Although simple, Bag of Words is highly effective for baseline text classification tasks.

5. Model Development

A **Random Forest Classifier** was selected for sentiment classification. Random Forest is an ensemble learning method that builds multiple decision trees and combines their outputs to improve accuracy and reduce overfitting.

Key model settings:

- Number of trees (estimators): 100
- Criterion: Gini impurity (default)
- Train-test split: 80% training, 20% testing

The model was trained using the processed training dataset and evaluated using the test dataset.

Random Forest was chosen due to its robustness, ability to handle high-dimensional data, and strong performance on classification tasks.

6. Model Evaluation

The performance of the model was evaluated using multiple classification metrics:

- **Accuracy**: Measures the overall correctness of predictions
- **Precision**: Measures how many predicted positive cases were actually positive
- **Recall**: Measures how many actual positive cases were correctly identified
- **F1-score**: Harmonic mean of precision and recall

Additionally, a **confusion matrix** was used to visualize prediction outcomes, including true positives, true negatives, false positives, and false negatives.

The results indicate that the model performs well in classifying sentiment, with strong accuracy and balanced precision-recall values. This suggests that the model is effective at distinguishing between positive and negative reviews.

7. Results and Discussion

The trained Random Forest model achieved high performance on both training and testing datasets. The similarity between training and testing accuracy suggests that the model generalizes well and does not suffer significantly from overfitting.

Feature importance analysis revealed that certain words strongly influence sentiment classification. Words such as “love”, “great”, and “excellent” were strongly associated with positive sentiment, while words like “bad”, “poor”, and “disappointed” were associated with negative sentiment.

This indicates that the model successfully captures meaningful linguistic patterns from customer reviews.

8. Deployment Strategy

To deploy this sentiment analysis model in a real-world environment, the following steps can be followed:

1. Save the trained model using serialization tools such as joblib or pickle
2. Develop a REST API using Flask or FastAPI to serve predictions
3. Deploy the API on cloud platforms such as AWS, Azure, or Heroku
4. Integrate the model into business applications such as:
    - E-commerce review systems
    - Customer feedback dashboards
    - Chatbots and virtual assistants

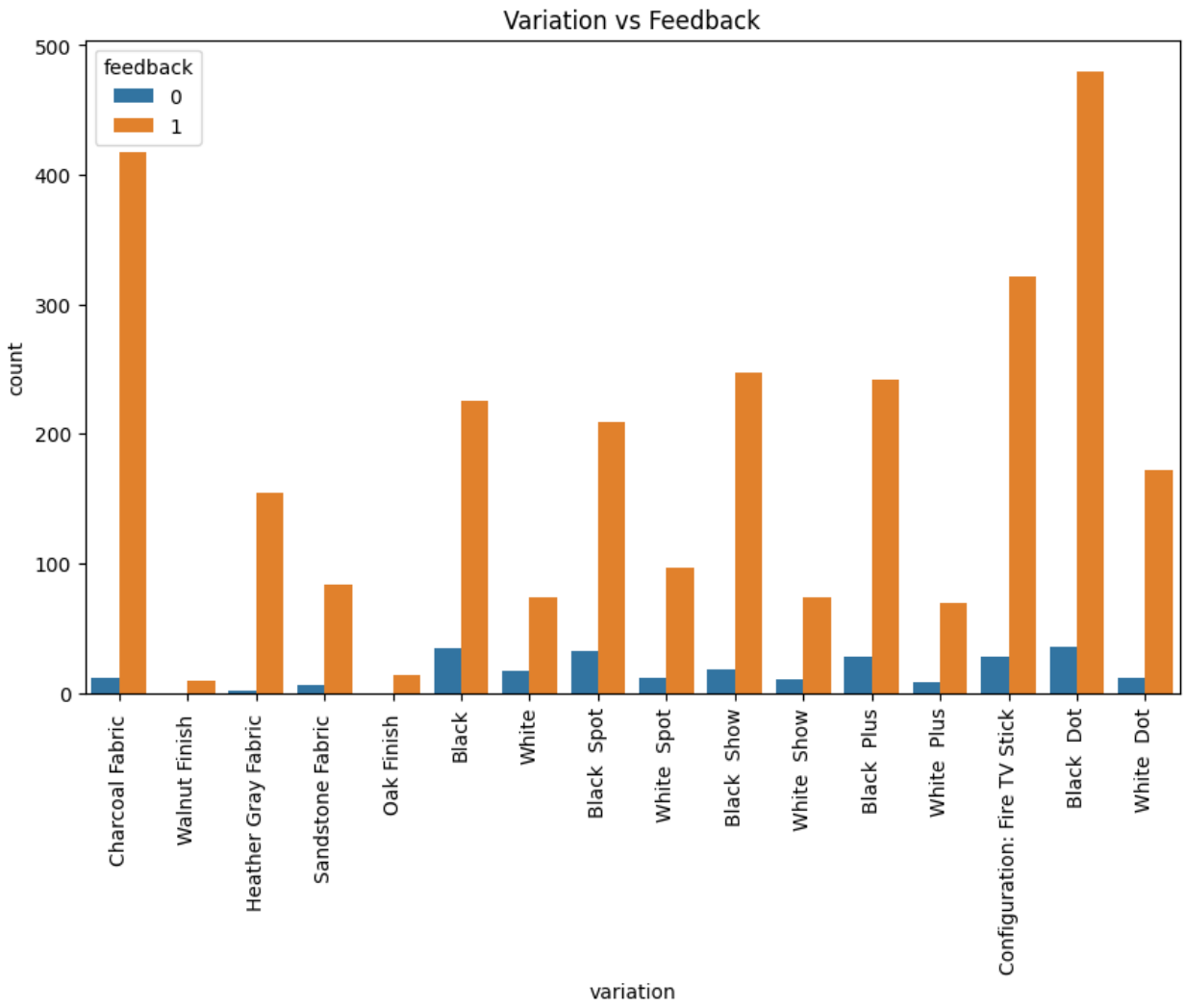
This enables real-time sentiment classification of incoming customer reviews.

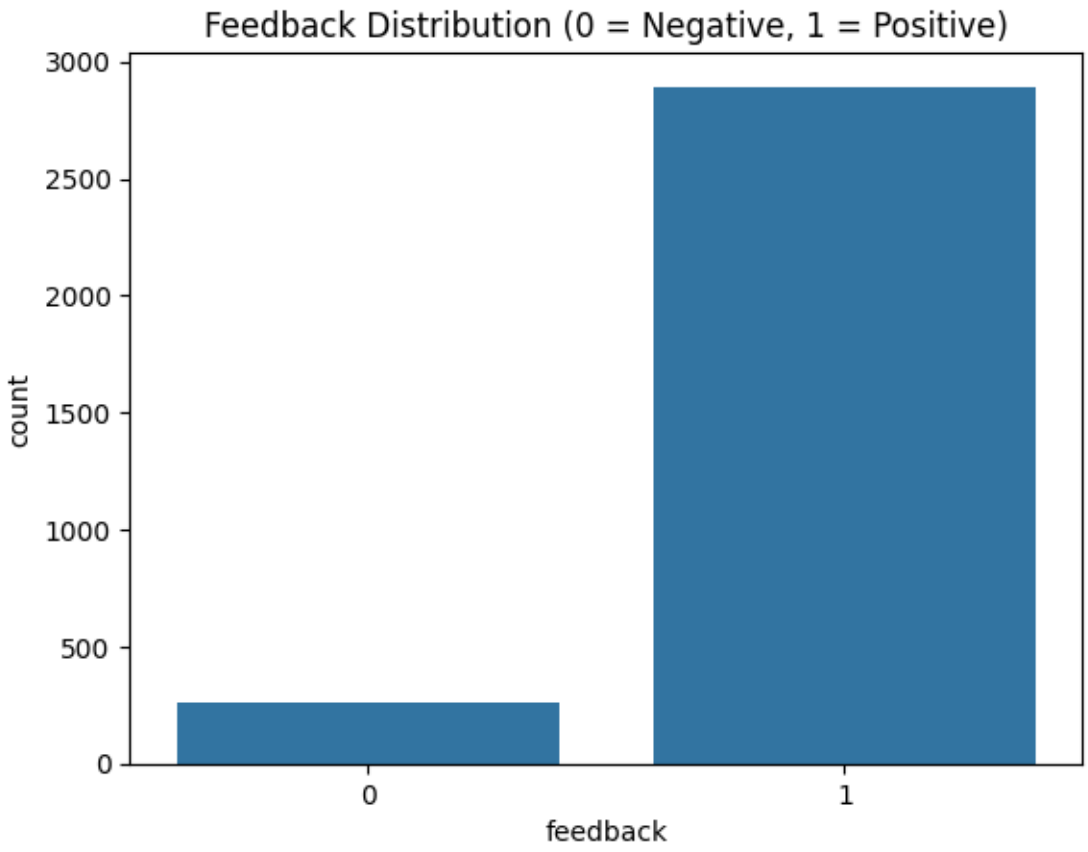
9. Conclusion

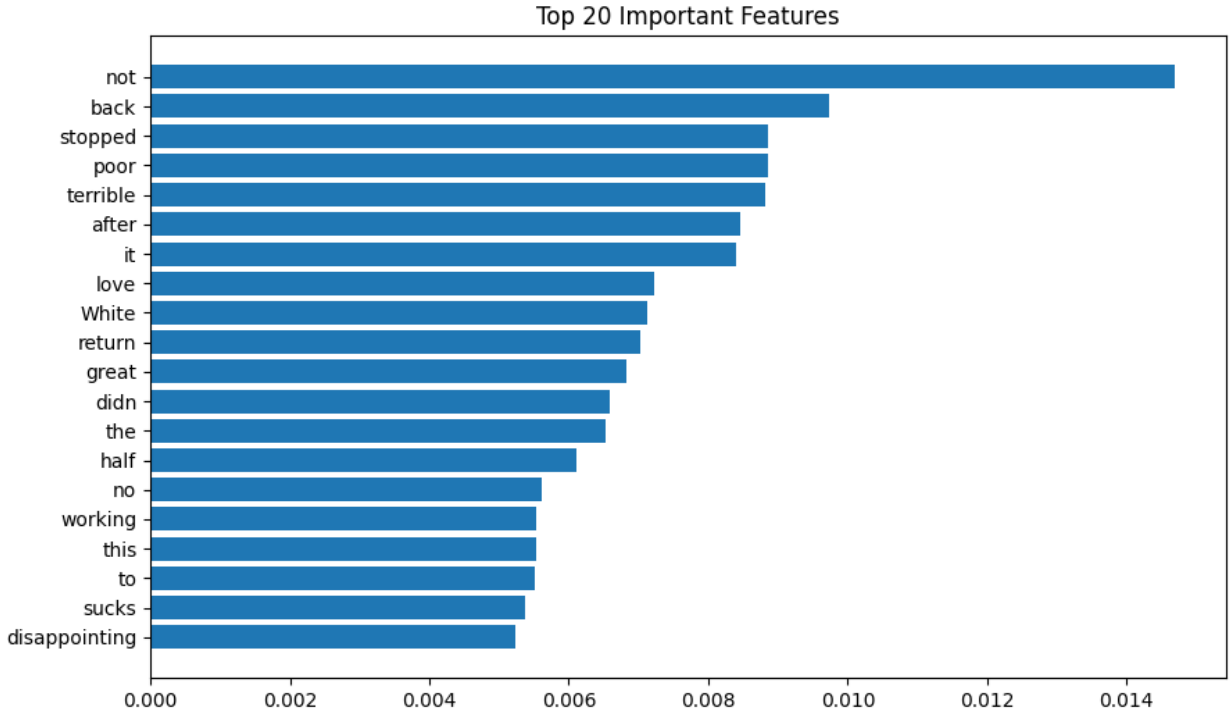
This project successfully demonstrates the application of machine learning techniques for sentiment analysis of Amazon Alexa customer reviews. By using text preprocessing, feature engineering, and a Random Forest classifier, the model effectively classifies reviews into positive and negative categories.

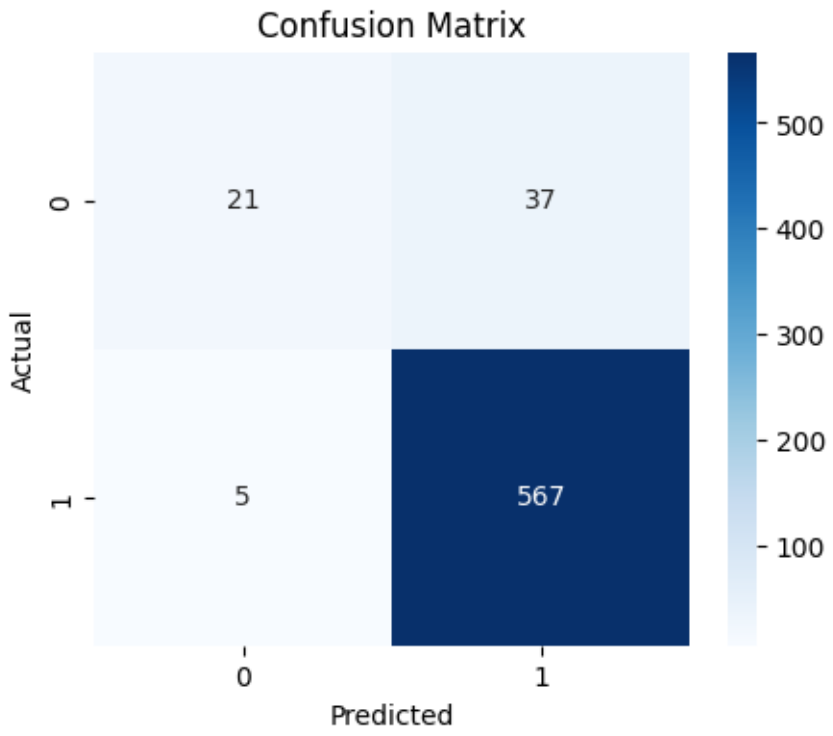
The results show that machine learning can provide valuable insights into customer sentiment and help organizations better understand user feedback.

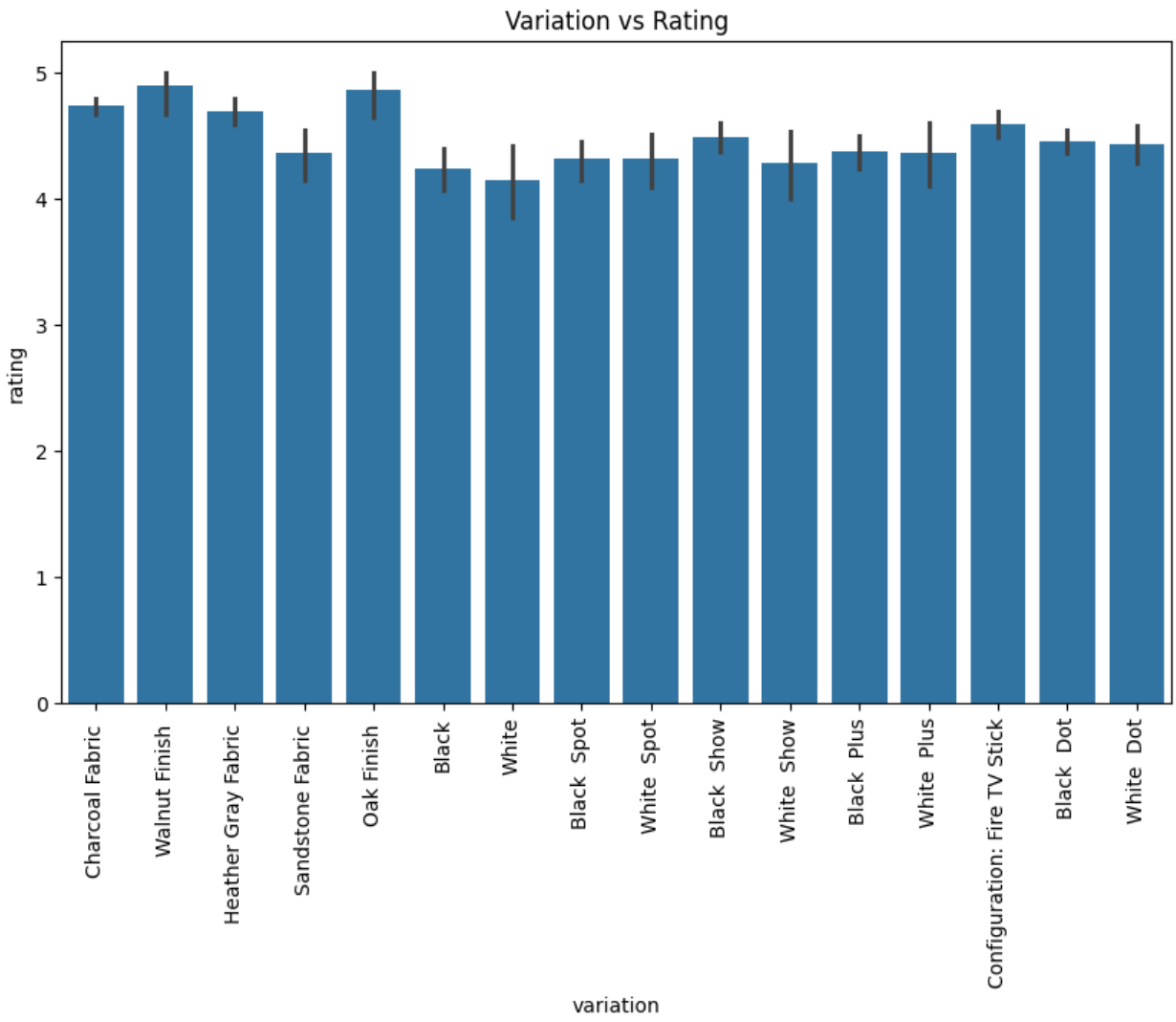
Future improvements could include the use of TF-IDF vectorization, deep learning models such as LSTM, or transformer-based models like BERT to further enhance performance and accuracy.

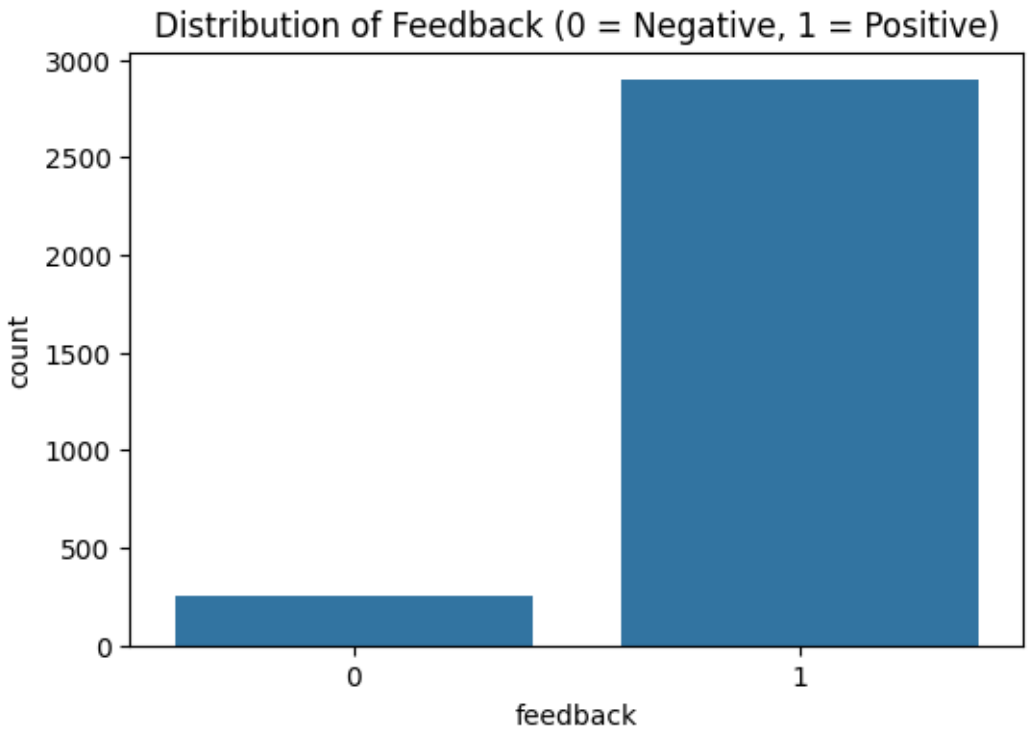

Variation vs Feedback
feedback
0
1
count
500
400
300
200
100
0
Charcoal Fabric
Walnut Finish
Heather Gray Fabric
Sandstone Fabric
Oak Finish
Black
White
Black Spot
White Spot
Black Show
White Show
Black Plus
White Plus
Configuration: Fire TV Stick
Black Dot
White Dot
variation

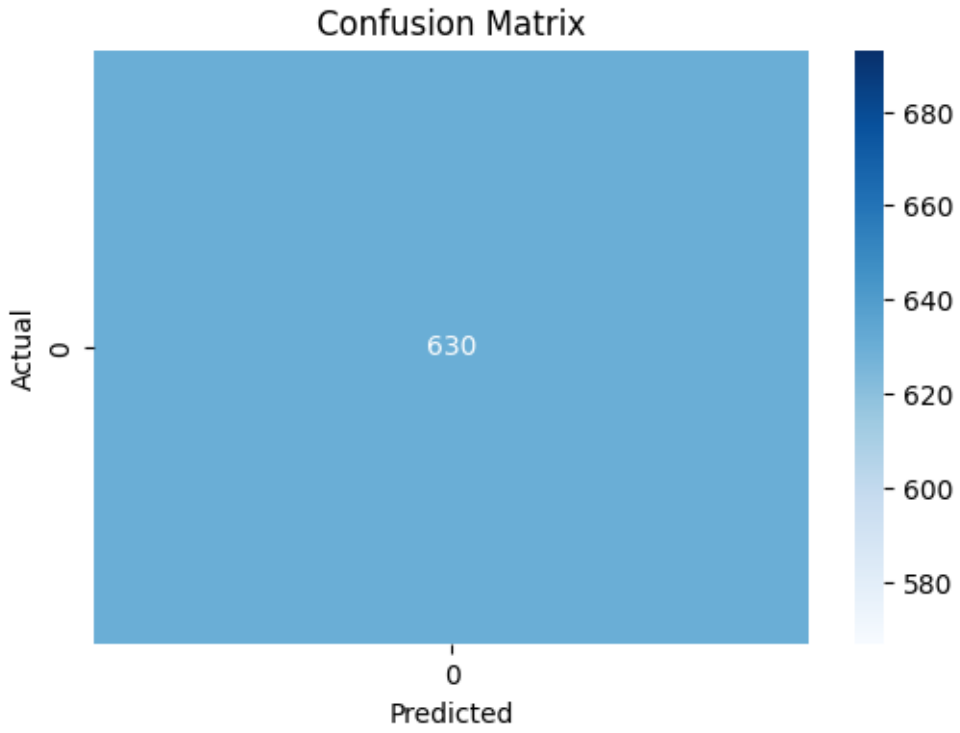

Feedback Distribution (0 = Negative, 1 = Positive)
count
feedback
0
1
0
500
1000
1500
2000
2500
3000


Top 20 Important Features
not
back
stopped
poor
terrible
after
it
love
White
return
great
didn
the
half
no
working
this
to
sucks
disappointing
0.000
0.002
0.004
0.006
0.008
0.010
0.012
0.014


Confusion Matrix
21
37
5
567
0
1
Actual
0
1
Predicted
500
400
300
200
100


Variation vs Rating
rating
0
1
2
3
4
5
Charcoal Fabric
Walnut Finish
Heather Gray Fabric
Sandstone Fabric
Oak Finish
Black
White
Black Spot
White Spot
Black Show
White Show
Black Plus
White Plus
Configuration: Fire TV Stick
Black Dot
White Dot
variation





GitHub repository Link :-

https://github.com/kadiwala1234/Customer-Review-Sentiment-Analysis